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RELATIONSHIP BETWEEN ARTIFICIAL INTELLIGENCE (AI) AND TEACHERS' EDUCATION PROGRAMMES FOR SUSTAINABLE NATIONAL DEVELOPMENT IN ENTREPRENEURSHIP EDUCATION IN PUBLIC SCHOOLS OF NORTH CENTRAL NIGERIA

Saleh Agwom Dauda, PhD, Ogah Felix Mathew, PhD, Adepoju Oluseye Adediran

ABSTRACT

This study investigated the relationship between Artificial Intelligence (AI) and teachers' education programmes for sustainable national development in entrepreneurship education in public schools of North Central Nigeria. The study focused on how AI-driven instructional tools, intelligent tutoring systems, and adaptive learning technologies enhance teacher preparation, pedagogical competence, and innovative practices in entrepreneurship education. A descriptive survey design was adopted, involving a population of 42,316 lecturers, teacher trainers, and students from Colleges of Education and Faculties of Education offering entrepreneurship-related programmes in the North Central zone. Using Krejcie and Morgan's sampling table, a sample size of 380 respondents was selected. A structured questionnaire validated by three experts in educational management and technology achieved a Content Validity Index (CVI) of 0.82 and a Cronbach Alpha reliability coefficient of 0.81. Data were analyzed using descriptive and inferential statistics at a 0.05 significance level. The findings revealed a significant positive relationship between AI integration and teachers' education programmes in enhancing entrepreneurship skills, creativity, problem-solving, and innovation among pre-service and in-service teachers. However, the study also identified major constraints such as inadequate ICT infrastructure, poor funding, low AI literacy among teacher educators, and limited institutional support. The study concludes that the effective integration of AI into entrepreneurship teacher education is essential for achieving sustainable national development by promoting self-reliance, job creation, and technological innovation. It recommends the inclusion of AI-based entrepreneurial pedagogy in teacher training curricula and continuous professional development for educators to strengthen digital competence and instructional efficiency.

Keywords: *Artificial Intelligence, Teachers' Education Programme, Entrepreneurship Education, Sustainable National Development, Public Schools*

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Relationship Between Artificial Intelligence (AI) And Teachers' Education Programmes For Sustainable National Development In Entrepreneurship Education In Public Schools Of North Central Nigeria

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Relationship Between Artificial Intelligence (AI) And Teachers' Education Programmes For Sustainable National Development In Entrepreneurship Education In Public Schools Of North Central Nigeria

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This study investigated the relationship between Artificial Intelligence (AI) and teachers' education programmes for sustainable national development in entrepreneurship education in public schools of North Central Nigeria. The study focused on how AI-driven instructional tools, intelligent tutoring systems, and adaptive learning technologies enhance teacher preparation, pedagogical competence, and innovative practices in entrepreneurship education. A descriptive survey design was adopted, involving a population of 42,316 lecturers, teacher trainers, and students from Colleges of Education and Faculties of Education offering entrepreneurship-related programmes in the North Central zone. Using Krejcie and Morgan's sampling table, a sample size of 380 respondents was selected. A structured questionnaire validated by three experts in educational management and technology achieved a Content Validity Index (CVI) of 0.82 and a Cronbach Alpha reliability coefficient of 0.81. Data were analyzed using descriptive and inferential statistics at a 0.05 significance level. The findings revealed a significant positive relationship between AI integration and teachers' education programmes in enhancing entrepreneurship skills, creativity, problem-solving, and innovation among pre-service and in-service teachers. However, major constraints such as inadequate ICT infrastructure, poor funding, low AI literacy among teacher educators, and limited institutional support were identified. The study concludes that effective AI integration into entrepreneurship teacher education is essential for sustainable national development, promoting self-reliance, job creation, and technological innovation. It recommends the inclusion of AI-based entrepreneurial pedagogy in teacher training curricula and continuous professional development for educators to strengthen digital competence and instructional efficiency.

Keywords

Artificial Intelligence, Teachers' Education Programme, Entrepreneurship Education, Sustainable National Development, Public Schools, North Central Nigeria

INTRODUCTION

Education is the cornerstone of human advancement, acting as the most potent tool for social transformation, economic growth, and sustainable national development. In the 21st century, the integration of technology-particularly Artificial Intelligence (AI)-has redefined how knowledge is created, shared, and applied. Artificial Intelligence, as defined by Russell and Norvig (2021), simulates human intelligence in machines designed for reasoning, learning, and problem-solving...

Statement of the Problem

Teacher education is the foundation upon which the quality and sustainability of a nation's educational system are built...

Research Questions

What is the relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

What is the relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

What opportunities does Artificial Intelligence offer for improving teachers' education programmes in basic education in public schools of North Central Nigeria?

What challenges hinder the effective integration of Artificial Intelligence into teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria?

Hypotheses

There is no significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the opportunities for improving teachers' education programmes in basic education in public schools of North Central Nigeria.

There is no significant relationship between Artificial Intelligence and the challenges that hinder effective teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria.

METHODOLOGY

The study adopted a descriptive survey research design, considered appropriate for analyzing existing conditions and relationships without manipulating the variables under investigation...

RESULTS

Research Question 1

What is the relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

Item No
Item Description
SA
A

D
SD
X?
SD
Decision

1
AI enhances instructional delivery and pedagogical innovation in the NCE basic education programme.
168
129
42
18
3.25
0.84
Agreed

2
Integration of AI tools improves teacher trainees' digital literacy and pedagogical competence for sustainable teaching.
162
132
45
16
3.23
0.83
Agreed

3
AI supports adaptive learning systems that promote individualized instruction in NCE basic education.
159
135
44
19
3.21
0.85
Agreed

4
AI-based assessment tools enhance evaluation of students' learning outcomes in NCE programmes.
157
137
45
18
3.20
0.84
Agreed

5
AI contributes to achieving quality assurance and sustainability in basic teacher education.
160
134
43
19
3.21

0.83
Agreed

Cluster Mean
3.22
0.84
Agreed

Research Question 2

What is the relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education in public schools of North Central Nigeria?

Item No
Item Description
SA
A
D
SD
X?
SD
Decision

6
AI promotes efficient supervision and mentorship of PDE teaching practice.
163
133
41
17
3.26
0.82
Agreed

7
AI applications enhance instructional planning and content delivery for PDE students.
158
136
43
17
3.23
0.83
Agreed

8
AI supports skill-based and continuous assessment for PDE trainees.
152
139
44
19
3.19
0.84
Agreed

9

AI facilitates flexible and virtual teaching environments for PDE trainees in basic education.
155
136
45
18
3.21
0.85
Agreed

10
AI improves professional competence and readiness of PDE graduates for sustainable education.
159
133
43
17
3.23
0.84
Agreed

Cluster Mean
3.22
0.84
Agreed

Research Question 3

What opportunities does Artificial Intelligence offer for improving teachers' education programmes in basic education in public schools of North Central Nigeria?

Item No
Item Description
SA
A
D
SD
X?
SD
Decision

11
AI enhances instructional efficiency and reduces administrative workload in teacher education.
160
137
43
18
3.23
0.83
Agreed

12
AI provides real-time feedback for trainee teachers during classroom micro-teaching.
157
136
45

19
3.21
0.84
Agreed

13
AI supports the creation of digital teaching aids and lesson resources.
153
139
46
18
3.19
0.85
Agreed

14
AI promotes equal access to modern instructional resources in teacher training institutions.
155
137
44
19
3.20
0.84
Agreed

15
AI fosters continuous professional development through intelligent tutoring systems.
160
133
45
17
3.23
0.82
Agreed

Cluster Mean
3.21
0.84
Agreed

Research Question 4

What challenges hinder the effective integration of Artificial Intelligence into teachers' education programmes for sustainable national development in basic education in public schools of North Central Nigeria?

Item No
Item Description
SA
A
D
SD
X?
SD
Decision

16
Inadequate ICT and digital infrastructure limits AI integration in basic education institutions.
165
131
41
17
3.27
0.82
Agreed

17
Low AI literacy among teacher educators hinders adoption.
162
132
44
18
3.23
0.84
Agreed

18
Insufficient funding for digital innovation constrains AI adoption.
155
136
46
19
3.20
0.85
Agreed

19
Lack of clear policy and curriculum framework for AI integration in teacher education.
153
138
44
19
3.19
0.84
Agreed

20
Resistance to technological change among educators.
150
140
45
20
3.18
0.85
Agreed

Cluster Mean
3.21
0.84

Agreed

Test of Hypotheses

Test of Null Hypothesis 1 (Ho?)

There is no significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education.

Responses
Observed Freq
Expected Freq
??-cal
??-tab
Df
p-value
Alpha
Decision

Agreed
297
351
228.41
13.067
350
0.011
0.05
Reject

Disagreed
55

Interpretation: Since ??-cal (228.41) > ??-tab (13.067) and p-value (0.011) < 0.05, the null hypothesis is rejected. Hence, there is a significant relationship between Artificial Intelligence and the NCE programme for sustainable national development in basic education.

Test of Null Hypothesis 2 (Ho?)

There is no significant relationship between Artificial Intelligence and the Professional Diploma in Education (PDE) programme for sustainable national development in basic education.

Responses
Observed Freq
Expected Freq
??-cal
??-tab
Df
p-value
Alpha
Decision

Agreed
290
351
231.65
13.067

350
0.014
0.05
Reject

Disagreed
61

Interpretation: Because ??-cal (231.65) > ??-tab (13.067) and p-value (0.014) < 0.05, the null hypothesis is rejected. Therefore, there is a significant relationship between Artificial Intelligence and the PDE programme for sustainable national development in basic education.

Test of Null Hypothesis 3 (Ho?)

There is no significant relationship between Artificial Intelligence and opportunities for improving teachers' education programmes in basic education.

Responses
Observed Freq
Expected Freq
??-cal
??-tab
Df
p-value
Alpha
Decision

Agreed
298
351
233.52
13.067
350
0.010
0.05
Reject

Disagreed
53

Interpretation: The ??-calculated value (233.52) is greater than the table value (13.067) and the p-value (0.010) is less than 0.05. Therefore, the null hypothesis is rejected, confirming that AI offers significant opportunities for improving teachers' education programmes in basic education.

Test of Null Hypothesis 4 (Ho?)

There is no significant relationship between Artificial Intelligence and the challenges that hinder effective teachers' education programmes for sustainable national development in basic education.

Responses
Observed Freq
Expected Freq
??-cal
??-tab
Df
p-value

Alpha
Decision

Agreed
294
351
229.84
13.067
350
0.018
0.05
Reject

Disagreed
58

Interpretation: Since ??-calculated (229.84) > ??-tab (13.067) and p (0.018) < 0.05, the null hypothesis is rejected. This indicates a significant relationship between AI integration and the challenges affecting teachers' education programmes in basic education in North Central Nigeria.

DISCUSSION OF FINDINGS

The first major finding indicated a significant relationship between Artificial Intelligence and the National Certificate in Education (NCE) programme for sustainable national development in basic education...

CONCLUSION

The study examined the relationship between Artificial Intelligence (AI) and teachers' education programmes for sustainable national development in basic education within public schools of North Central Nigeria...

RECOMMENDATIONS

The National Commission for Colleges of Education (NCCE) and teacher education faculties should incorporate Artificial Intelligence literacy, machine learning concepts, and digital pedagogy into NCE and PDE curricula...

Regular training, workshops, and certification programmes should be organized to enhance lecturers' and teacher trainers' proficiency in AI applications...

The Federal and State Ministries of Education should increase funding allocations for ICT infrastructure in teacher training institutions...

A comprehensive policy framework should be developed to guide the integration of AI into teacher education programmes...

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